

CONGENITAL HEART DISEASE — Revision Table

Bold = high-yield. 8/1000 liveborn; VSD commonest. [bracket] = standard added.

ACYANOTIC (left-to-right shunt)			
Lesion	Signs / murmur	Investigations	Treatment
ASD (secundum commonest; primum = partial AVSD)	Wide & FIXED split S2, soft ejection systolic; primum → apical pansystolic (AV regurg)	CXR: RA/RV enlarge, ↑pulmonary. ECG: secundum = RBBB + RAD; AVSD = SUPERIOR axis. Echo mainstay	Secundum → device closure; primum/sinus venosus → surgery (age 3–5yr)
VSD (commonest CHD, 30%)	Small = loud pansystolic LLSE (loud = small), quiet P2. Large = soft/no murmur, loud P2, mid-diastolic apical, HF	CXR: large = cardiomegaly + plethora. ECG: large = biventricular hypertrophy. Echo dx	Small → reassure + IE prophylaxis. Large → diuretic + ACEi + surgical closure
PDA (6–8%, F:M 2:1, prematures)	Machinery continuous murmur (2nd L ICS), bounding pulses, wide pulse pressure, thrill	CXR: LV/LA cardiomegaly. ECG: LV/biventricular hypertrophy. Echo	Transcatheter closure / ligation (<1yr); [preterm = indomethacin/ibuprofen]. IE risk
CYANOTIC			
Lesion	Key features	Investigations	Treatment
TGA (d-TGA)	5% CHD; aorta from RV. Cyanosis ALWAYS, presents day 2 (ductal closure); loud single S2. Survival needs mixing (PFO/PDA/VSD). Diabetic mother; M:F 3:1	CXR: egg-shaped heart, narrow mediastinum. ECG normal neonatal RV dominance. Echo	PGE infusion (keep duct open) + balloon atrial septostomy (Rashkind) + arterial switch (Jatene, first 2 wks)
TOF (5–7%)	4: PS (RVOT obstruction) + VSD + overriding aorta + RVH. Cyanosis (severe = PDA-dependent), clubbing, squatting. Tet spells (murmur DECREASES), first 2 yrs	CXR: boot-shaped heart, oligoemic lungs, R aortic arch 20–25%. ECG: RVH, RAD, P pulmonale	Medical → definitive repair ~6mo (close VSD + relieve RVOT). Cyanosed neonate → modified Blalock-Taussig shunt. Spells: knee-chest + morphine + IV propranolol + fluids + bicarbonate